MORPHISM PROPERTIES (FINITE, CLOSED IMMERSION, FLAT, FINITE TYPE, FP, ETALE) TRANSPORT ACROSS THE GLUEING EQUIVALENCE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2581-2589 of sources/stacks-spaces-pushouts/pinned-original-source.tex; proof span is lines 2591-2645, the complete author proof (53 body lines).

Editorial note. Navigation stubs: 7 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 8 of queue_spaces-pushouts.json (pb=2734, stmt 2581-2589, proof 2591-2645).

Editorial note. spaces-pushouts wave-1: fresh adjudication accept (2026-10-14); six-check dispatched pool (§2 row appended 02:4x).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 0F9Q: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2547–2555. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BNL: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2645–2645. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BP7: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2645–2645. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BNN: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2645–2645. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02GV: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2645–2645. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02G8: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2645–2645. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 02G7: sources/stacks-spaces-pushouts/pinned-original-source.tex, source lines 2645–2645. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let P be one of the following properties of morphisms: "finite", "closed immersion", "flat", "finite type", "flat and finite presentation", "étale". Under the equivalence of Lemma 0F9Q the morphisms having P correspond to morphisms of triples whose components have P.*

Proof. Let P' be one of the following properties of homomorphisms of rings: "finite", "surjective", "flat", "finite type", "flat and of finite presentation", "étale". Translated into algebra, the statement means the following: If $A \to B$ is an R-algebra homomorphism and A and B are glueable for $(R \to R', f)$, then $A_f \to B_f$ and $A \otimes_R R' \to B \otimes_R R'$ have P' if and only if $A \to B$ has P'.

By More on Algebra, Lemmas 0BNN and 0BP7 the algebraic statement is true for P' equal to "finite" or "flat".

If $A_f \to B_f$ and $A \otimes_R R' \to B \otimes_R R'$ are surjective, then $N = B/A$ is an R-module with $N_f = 0$ and $N \otimes_R R' = 0$ and hence vanishes by More on Algebra, Lemma 0BNL. Thus $A \to B$ is surjective.

If $A_f \to B_f$ and $A \otimes_R R' \to B \otimes_R R'$ are finite type, then we can choose an A-algebra homomorphism $A[x_1, \ldots, x_n] \to B$ such that $A_f[x_1, \ldots, x_n] \to B_f$ and $(A \otimes_R R')[x_1, \ldots, x_n] \to B \otimes_R R'$ are surjective (small detail omitted). We conclude that $A[x_1, \ldots, x_n] \to B$ is surjective by the previous result. Thus $A \to B$ is of finite type.

If $A_f \to B_f$ and $A \otimes_R R' \to B \otimes_R R'$ are flat and of finite presentation, then we know that $A \to B$ is flat and of finite type by what we have already shown. Choose a surjection $A[x_1, \ldots, x_n] \to B$ and denote I the kernel. By flatness of B over A we see that I_f is the kernel of $A_f[x_1, \ldots, x_n] \to B_f$ and $I \otimes_R R'$ is the kernel of $A \otimes_R R'[x_1, \ldots, x_n] \to B \otimes_R R'$. Thus I_f is a finite $A_f[x_1, \ldots, x_n]$-module and $I \otimes_R R'$ is a finite $(A \otimes_R R')[x_1, \ldots, x_n]$-module. By More on Algebra, Lemma 0BNN applied to I viewed as a module over $A[x_1, \ldots, x_n]$ we conclude that I is a finitely generated ideal and we conclude $A \to B$ is flat and of finite presentation.

If $A_f \to B_f$ and $A \otimes_R R' \to B \otimes_R R'$ are étale, then we know that $A \to B$ is flat and of finite presentation by what we have already shown. Since the fibres of $\mathrm{Spec}(B) \to \mathrm{Spec}(A)$ are isomorphic to fibres of $\mathrm{Spec}(B_f) \to \mathrm{Spec}(A_f)$ or $\mathrm{Spec}(B/fB) \to \mathrm{Spec}(A/fA)$, we conclude that $A \to B$ is unramified, see Morphisms, Lemmas 02G7 and 02G8. We conclude that $A \to B$ is étale by Morphisms, Lemma 02GV for example. □